Chapter 5 : Probability Theory

12 June 2024 09:41

⇒ If A and B are disjoint events, then

$P(A \text{ or } B) = P(A) + P(B)$ } addition rule

↳ the probability that one or the other of two events A and B occurs when A and B cannot occur together.

Q) Is it same as independent events ?

A) No, events can also be independent & not disjoint

ex:- X : heads on first toss

Y : heads on second toss

So if the tosses results in HH, both X & Y have occurred together.

Also in disjoint events the fact A occurs implies B cannot occur or viceversa says that A & B are not independent, because the information that A occurred changed the probability of B.

⇒ Independence means the output of the first toss cannot influence the outcomes of the second toss.

EXAMPLE 5.1 Independent or Not?

Because a coin has no memory and most coin tossers cannot influence the fall of the coin, it is safe to assume that successive coin tosses are independent. For a balanced coin this means that, after we see the outcome of the first toss, we still assign probability 1/2 to heads on the second toss.

On the other hand, the colors of successive cards dealt from the same deck are not independent. A standard 52-card deck contains 26 red and 26 black cards. For the first card dealt from a shuffled deck, the probability of a red card is $26/52 = 0.50$ (equally likely outcomes). Once we see that the first card is red, we know that there are only 25 reds among the remaining 51 cards. The probability that the second card is red is therefore only $25/51 = 0.49$. Knowing the outcome of the first deal changes the probabilities for the second.

If two individuals request a credit report from a particular credit agency (for example, Equifax, TransUnion, or Experian), it is reasonable to assume that the two credit scores are independent because the credit history of one customer does not influence the credit history of the other customer. But if one individual were to request a credit report in two successive years, then the two credit scores would likely not be independent. Knowing the initial credit score might lead the individual to try to improve his credit record which would influence the subsequent credit score.

- Multiplication rule for independent events.

⇒ Two events A and B are independent if knowing that one occurs does not change the probability that other occurs.

∴ If A & B occurs

$P(A \text{ and } B) = P(A)\,P(B)$

⇒ If the above equality doesn't hold true, then A & B are not independent events.

EXAMPLE 5.2 Determining Independence Using the Multiplication Rule

Consider a manufacturer that uses two suppliers for supplying an identical part that enters the production line. Sixty percent of the parts come from one supplier, while the remaining 40% come from the other supplier. Internal quality audits find that there is a 1% chance that a randomly chosen part from the production line is defective. External supplier audits reveal that two parts per thousand are defective from Supplier 1. Are the events of a part coming from a particular supplier, say Supplier 1, and a part being defective independent?

Define the two events as follows:

$S1$ = A randomly chosen part comes from Supplier 1.

D = A randomly chosen part is defective.

We have $P(S1) = 0.60$ and $P(D) = 0.01$. The product of these probabilities is

$$P(S1)P(D) = (0.60)(0.01) = 0.006$$

However, supplier audits of Supplier 1 indicate that $P(S1 \text{ and } D) = 0.002$. Given that $P(S1 \text{ and } D) \neq P(S1)P(D)$, we conclude that the supplier and defective part events are not independent.

- General addition rule :

P(A or B) = P(A) + P(B) - P(A and B)

or

P(A + B) = P(A) + P(B) - P(A and B).

If A & B are disjoint then P(A and B) = 0.

- General Multiplication rule for any two events :

⇒ The probability that two events can happen together can be found by.

P(A and B) = P(A|B) · P(B) = P(B|A) · P(A).

⇒ Where P(B|A) is the conditional probability that B occurs, given the information that A occurs.

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CHAPTER 5 Probability Theory

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EXAMPLE 5.9 Career in Big Business: NFL

Worldwide, the sports industry has become synonymous with big business. It has been estimated by the United Nations that sports account for nearly 3% of global economic activity.[4] The most profitable sport in the world is professional football under the management of the National Football League (NFL).[5] With multimillion-dollar signing contracts, the economic appeal of pursuing a career as a professional sports athlete is unquestionably strong. But what are the realities? Only 5.7% of high school football players go on to play at the college level. Of these, only 1.8% will play in the NFL.[6] About 40% of the NFL players have a career of more than 3 years. Define these events for the sport of football:

A = {competes in college}

B = {competes in the NFL}

C = {has an NFL career longer than 3 years}

What is the probability that a high school football player competes in college and then goes on to have an NFL career of more than 3 years? We know that

$$P(A) = 0.057$$
$$P(B \mid A) = 0.018$$
$$P(C \mid A \text{ and } B) = 0.4$$

The probability we want is therefore

$$P(A \text{ and } B \text{ and } C) = P(A)P(B \mid A)P(C \mid A \text{ and } B)$$
$$= 0.057 \times 0.018 \times 0.40 = 0.00041$$

Only about 4 of every 10,000 high school football players can expect to compete in college and have an NFL career of more than 3 years. High school football players would be wise to concentrate on studies rather than on unrealistic hopes of fortune from pro football. Hopes of fortune may be further tempered by the statistical fact that NFL players with careers longer than 3 years have an average life expectancy that is 20 years less than that of the typical U.S. male.

- Conditional Probability and independence :

won't make sense if A can never occur.

When $P(A) > 0$, the

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

- Independent events :-

If A and B are independent events then

$$P(B|A) = P(B)$$

Proof: $P(B|A) = \frac{P(B \text{ and } A)}{P(A)} \overset{\because \text{independence}}{\Longrightarrow} \frac{P(B)\,P(A)}{P(A)} = P(B)$.

- Tree diagrams & Bayes rule :

→ Probability often requires to combine several of the basic rules into a more elaborate calculation.

→ Here is an example that shows how to solve problems that have several stages.

:

- Bayes Rule

⇒ If A and B are any events whose probabilities are not 0 or 1.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B|A) \cdot P(A) + P(B|A^c) P(A^c)}$$

- The Binomial Distributions:

⇒ A company's human resources manager asks 100 employees if job stress is affecting their personal lives.
↳ How many will say yes?

⇒ A company develops 30 new products that are subject to market testing.
↳ How many products will be successful in market testing?

⇒ A store sells 60 computers with extended 3-year warranties.
↳ How many will not need repair within 3 years?

✱ In all three situations, we want probability model for a count of successful outcomes.

⇒ The distribution of a count depends on how the data are produced.

- The Binomial setting:

1) There are fixed number n of observations.

2) The n observations are all independent. That is, knowing the result of one observation tells you nothing about the other observations.

3 Each observation falls in one of the two categories, which for convience will call "success" and "failure".

4 The probability of success, call it p, is same for each observation.

⇒ Think of tossing coin n times as an example of the binomial setting.

- Binomial Distribution:

⇒ The distribution of the count X of successes in the binomial setting is the binomial distribution with parameters n and p.

↳ The parameter n is the number of observations, and p is the probability of a success on any one observation.

↳ Since X is the count of success, the possible values of X

are the whole numbers from 0 to n.

⇒ Not all counts have binomial distribution.

EXAMPLE 5.12 Determining Consumer Preferences

Market research to determine the product preferences of consumers is an increasingly important area in the intersection of business and statistics. ~~With some companies competing in markets with little product discrimination, determining what features consumers most prefer is critical to the success of a product.~~ Market research is interested in finding the probability of a "typical" consumer purchasing a product with a particular combination of features.

Suppose that your product is actually preferred over competitors' products by 25% of all consumers. If X is the count of the number of consumers who prefer your product in a group of 5 consumers, then X has a binomial distribution with $n = 5$ and $p = 0.25$, provided the 5 consumers make choices independently. Given their importance, the methods for collecting independent consumer data are part of most marketing curricula found in business schools. These methods are routinely employed by marketing research consulting firms, such as the Nielsen Company (www.nielsen.com) and Burke, Inc. (www.burke.com), as well as by in-house marketing research departments found in most major firms.

EXAMPLE 5.13 Dealing Cards (Not independent, hence not binomial)

Deal 10 cards from a shuffled deck and count the number X of red cards. There are 10 observations, and each gives either a red or a black card. A "success" is a red card. But the observations are *not* independent. If the first card is black, the second is more likely to be red because there are more red cards than black cards left in the deck. The count X does *not* have a binomial distribution.

D Case 5.1: Inspecting a Supplier's Product:

⇒ A manufacturing firm purchases component for its products from suppliers.

↳ There has been quality lapses in the switches supplied by a regular vendor.

⇒ While working with the supplier to improve its processes, the firm temporarily institutes an acceptance sampling plan to asses the quality of shipments of switches.

↳ If the random sample contains too many switches that don't conform to specifications, the firm will not accept the shipment.

⇒ An engineer at the firm chooses an SRS of 10 switches from the shipment of 10,000.

⇒ Suppose that 10% of switches in the shipment are nonconforming.

⇒ The engineer counts the number X of non conforming switches in the sample.

⇒ This is not a binomial setting.

↳ Just like the case of cards.

↳ Even if one switch is removed from the proportion then will end up changing the probability of defective chips for the others.

↳ So, if there are 1000 defective switch the proption remaining is 1000/9999 = 0.10001 if the first switch is OK.

↳ and will be 999/9999 = 0.9991 if the first switch fails.

↳ Hence the probability changes for second switch because of the knowledge of first, hence they're not independent.

⇒ But in practice the change is very small, hence the distribution of X is very close to binomial distribution with $n=10$ & $p=0.1$.

When the population is much larger than sample, a count of successes in an SRS of size n has approximately binomial distribution.

EXAMPLE 5.14 Determining Consumer Preferences

Each consumer has probability 0.25 of preferring your product over competitors' products. If we question 5 consumers, what is the probability that exactly 2 of them prefer your product?

The count of consumers preferring your product is a binomial random variable X with $n = 5$ tries and probability $p = 0.25$ of a success on each try. We want ~~$P(X = 2)$.~~

$P(X=2)$? for $n=5$ & $p=0.25$

⇒ Because the method doesn't depend on the specific example, let's use "S" for success & F for failures. Do the work in two steps.

Step 1. Find the probability that a specific 2 of the 5 trials, say first & 3rd day gives success. ∴ SFSFF, because the trials are independent.

$$P(SFSFF) = P(S)\,P(F)\,P(S)\,P(F)\,P(F)$$
$$= (0.25)(0.75)(0.25)(0.75)(0.75)$$
$$= (0.25)^2 (0.75)^3 .$$

Step 2: Observe that the probability of any arrangement of 2's & 3 F's has the same probability

→ Here are all possible arrangements.

SSFFF SFSFF SFFSF SFFFS FSSFF
FSFSF FSFFS FFSSF FFSFS FFFSS
} total 10 ways

∴ $P(X=2) = 10\,(0.25)^2\,(0.75)^3 = \underline{0.2637}$.

⇒ This pattern works for any arrangement of k-successes in n observations. We use the following fact to do the counting

◦ Binomial Coefficient :-

⇒ The number of ways of arranging k successes among n observations is given by binomial coefficient

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

(n choose k)

for $k = 0, 1, 2, \ldots, n$.

◦ Binomial Probability :-

⇒ If X has a binomial distribution with n observations and probability p of success and the possible value of X are $0, 1, 2, \ldots, n$. If k is any one of these values then,

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

EXAMPLE 5.15 Inspecting Switches

CASE 5.1 The number X of switches that fail inspection in Case 5.1 closely follows the binomial distribution with $n = 10$ and $p = 0.1$.

The probability that no more than 1 switch fails is

$$\begin{aligned} P(X \le 1) &= P(X=1) + P(X=0) \\ &= \binom{10}{1}(0.1)^1(0.9)^9 + \binom{10}{0}(0.1)^0(0.9)^{10} \\ &= \frac{10!}{1!9!}(0.1)(0.3874) + \frac{10!}{0!10!}(1)(0.3487) \\ &= (10)(0.1)(0.3874) + (1)(1)(0.3487) \\ &= 0.3874 + 0.3487 = 0.7361 \end{aligned}$$

This calculation uses the facts that $0! = 1$ and that $a^0 = 1$ for any number a other than 0. We see that about 74% of all samples will contain no more than 1 bad switch. In fact, 35% of the samples will contain no bad switches. A sample of size 10 cannot be trusted to alert the engineer to the presence of unacceptable items in the shipment. Calculations such as this are used to design acceptance sampling schemes.

- Binomial mean and standard deviation :-

⇒ μ = average count of successes.

⇒ If the count X has binomial distribution with number of observations n and probability of success p, the

mean $\mu = np$.

$\sigma = \sqrt{np(1-p)}$

EXAMPLE 5.18 Inspecting Switches

CASE 5.1 Continuing Case 5.1, the count X of bad switches is binomial with $n = 10$ and $p = 0.1$. The mean and standard deviation of this binomial distribution are

$$\begin{aligned} \mu &= np \\ &= (10)(0.1) = 1 \\ \sigma &= \sqrt{np(1-p)} \\ &= \sqrt{(10)(0.1)(0.9)} = \sqrt{0.9} = 0.9487 \end{aligned}$$

In Figure 5.8, we have added the mean to the probability histogram of the distribution.

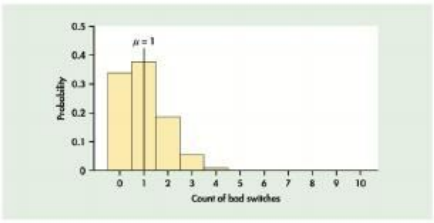
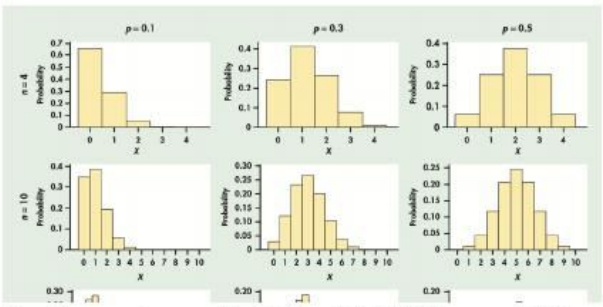


FIGURE 5.8 Probability histogram for the binomial distribution with $n = 10$ and $p = 0.1$ and $\mu = 1$ marked, for Example 5.18.

- The normal approximations to binomial distributions :

⇒ For a given p, the shape of the binomial distribution becomes more symmetrical as n gets larger.

⇒ Also for a given n the binomial distribution is more symmetrical as p approaches 0.5.



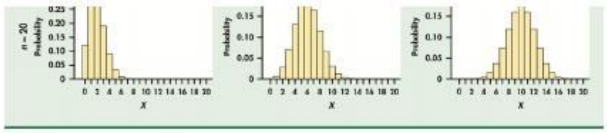


FIGURE 5.9 The shape of the binomial distribution for different values of n and p.

- Normal Approximation for binomial distribution.
  - Suppose that the count X has the binomial distribution with n trials and success probability p.
    - When n is large, the distribution of X is approximately Normal. $N(np, \sqrt{np(1-p)})$.
      - As a rule of thumb if $np \geq 10$ and $n(1-p) \geq 10$, we can use Normal approximation.

- The Poisson Distributions :-

⇒ Count the number of finished defects in sheet metal of a car or refrigerator; the count could go upto 0, 1, 2, 3, ... so on.

The Poisson setting :-

⇒ The Poisson distribution is another distribution for count random variables.
- Count the number of events (call them success) that occur in some fixed unit of measure such as an area of sheet metal, a period time, or length of a cable.

The poisson setting

1. The number of successes that occur in any unit of measure is independent of the number of successes that occur in any non overlapping unit of measure.
2. The probability that a success will occur in a unit of measure is the same for all units of equal size, and is proportional to size of that unit.
3. The probability that 2 or more successes will occur in a unit approaches 0 as the size of the unit becomes smaller

⇒ The quantity important in specifying Poisson distributions is the mean number of successes occurring per unit of measure.

Binomial vs. Poisson assumptions

Aspect	Binomial model
What you observe	# of successes in n fixed, discrete trials
Basic unit	A "trial" with two outcomes: success/failure
Independence	Trials are independent of each other
Parameter held constant	Success probability per trial = (same for every trial)
Mean & variance	Mean = np, Var = np(1−p)
Typical use-case	Repeated experiments, survey quality checks

Connection:
If you take a binomial with very many trials (n → large small) such that np = λ stays constant, the binomial c Poisson is often used as an approximation when n is l

Rule of thumb:
- Known number of opportunities/trials? → Binomi
- Random number of opportunities over a continu Poisson.

- Poisson Distribution :

⇒ The distribution of the count X of success in poisson setting is the Poisson distribution with mean μ.
- The parameter μ is the mean number of successes per unit of measure.

↳ The possible values of X are the whole numbers 0, 1, 2 ... If k is a whole number 0 or greater, then

$$P(X=k) = \frac{e^{-\mu}\mu^k}{k!}$$

The standard deviation of the distribution is $\sqrt{\mu}$.

EXAMPLE 5.21 Flaws in Carpets

A carpet manufacturer knows that the number of flaws per square yard in a type of carpet material ~~varies with an average of 1.6 flaws per square yard~~. The count X of flaws per square yard can be modeled by the Poisson distribution with ~~$\mu = 1.6$~~. The unit of measure is a square yard of carpet material. What is the probability of no more than 2 defects in a randomly chosen square yard of this material?

We will calculate $P(X \le 2)$ in two ways:

1. Software can do the calculation:

	A	B	C
1	Poisson(0,1.6,0)=	0.2019	
2	Poisson(1,1.6,0)=	0.3230	
3	Poisson(2,1.6,0)=	0.2584	
4	SUM(B1:B3)=	0.7834	
5			

2. We can use the Poisson probability formula:

$$\begin{aligned} P(X \le 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= \frac{e^{-1.6}(1.6)^0}{0!} + \frac{e^{-1.6}(1.6)^1}{1!} + \frac{e^{-1.6}(1.6)^2}{2!} \\ &= 0.2019 + 0.3230 + 0.2584 = 0.7833 \end{aligned}$$

The software answer and the hand calculation differ by 0.0001 due to roundoff error in the hand calculation. The software calculates the individual probabilities to many significant digits even though it displays only four significant digits.

- The poisson Model :-

⇒ If we add counts of successes in non overlapping areas of space or time, we're just counting the successes in a larger area.

⇒ If X is a poisson random variable with μ_X and Y is a poisson random variable with mean μ_Y, and Y is independent of X, then X+Y is a random Poisson variable with mean $\mu_X + \mu_Y$.

EXAMPLE 5.23 Paint Finish Flaws

Auto bodies are painted during manufacture by robots programmed to move in such a way that the paint is uniform in thickness and quality. You are testing a newly programmed robot by counting paint sags caused by small areas receiving too much paint. Sags are more common on vertical surfaces. Suppose that counts of sags on the roof follow the Poisson model with mean 0.7 sags

per square yard and that counts on the side panels of the auto body follow the Poisson model with mean 1.4 sags per square yard. Counts in nonoverlapping areas are independent. Then

- The number of sags in 2 square yards of roof is a Poisson random variable with mean $0.7 + 0.7 = 1.4$.
- The total roof area of the auto body is 4.8 square yards. The number of paint sags on a roof is a Poisson random variable with mean $4.8 \times 0.7 = 3.36$.
- A square foot is 1/9 square yard. The number of paint sags in a square foot of roof is a Poisson random variable with mean $1/9 \times 0.7 = 0.078$.
- If we examine 1 square yard of roof and 1 square yard of side panel, the number of sags is a Poisson random variable with mean $0.7 + 1.4 = 2.1$.

4 different random variables possible because counts in non overlapping areas are independent.

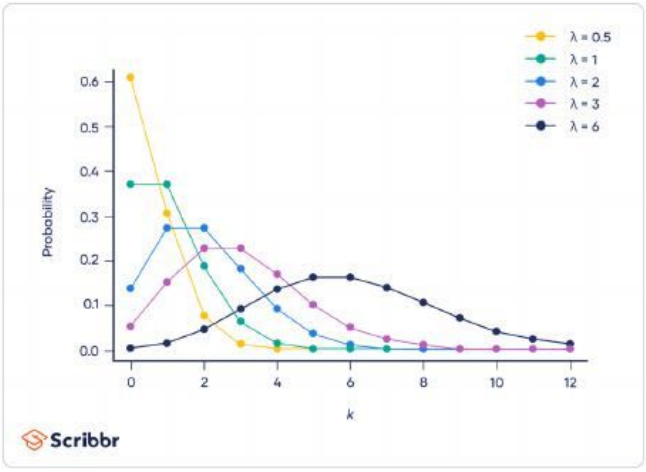
- More Distribution Approximations :

⇒ We've seen that the Normal distribution could be used to calculate binomial probabilities when n, the number of trials is large.

- Normal approximation to Poisson :

⇒ Some software is unable to calculate Poisson probabilities when μ is large.

↳ In such cases when μ is large, Poisson probabilities can be approximated using the Normal distribution with mean μ and the standard deviation $\sqrt{\mu}$.



- Poisson approximation to binomial :-

⇒ It's recommended to use normal approximation only when n & p satisfy $np \geq 10$ and $n(1-p) \geq 10$.

⇒ In cases p is so small such that $np < 10$, using Poisson with mean $\mu = np$ to calculate binomial probabilities yields more accurate results.

k	Binomial	Poisson
0	0.3677	0.3679
1	0.7358	0.7358
2	0.9198	0.9197
3	0.9811	0.9810
4	0.9964	0.9963

4	0.9964	0.9963
5	0.9994	0.9994
6	0.9999	0.9999
7	1.0000	1.0000